

DBMS Project - IT216

Milestone 1

Topic: Supply Chain Management System (SCMS)

Lab Group: 05

Team Members:

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1. Identify Database Application

Application Name: Supply Chain Management System (SCMS)

Domain: Operations, Procurement, and Logistics.

Business Context: We are modeling the operational backbone of a logistics company that handles both standard and temperature-sensitive goods (like pharmaceuticals or high-end perishables). The system bridges the gap between raw material procurement, warehouse storage, order fulfillment, and last-mile delivery.

2. Core Problem Statement and Solution Approach

The Problem: Modern supply chain operations frequently suffer from fragmented data management across procurement, inventory warehousing, and last-mile fulfillment. This fragmentation leads to costly inefficiencies: expired inventory due to poor stock rotation, fulfillment delays caused by inaccurate physical stock counts, and financial risks from untracked vendor support/reliability or customer credit defaults. Without a centralized, transaction-safe system, businesses cannot reliably enforce operational rules or maintain real-time visibility over physical goods.

The Solution: We propose the design and implementation of (an enterprise-grade) Supply Chain Management System (SCMS) database. This relational database will serve as the unified system of record for the entire product lifecycle. It is proposed to seamlessly integrate vendor cataloging, multi-zone inventory control, order processing, and shipment dispatch into a single, cohesive schema. By embedding core business logic—such as First-Expired-First-Out (FEFO) allocation, credit hold enforcement, and automated reorders directly into the database using constraints, triggers, and stored procedures, the system will guarantee data integrity, eliminate workflow inconsistencies, and provide a reliable foundation for operational decision-making. The system also supports inter-warehouse stock transfers, where inventory can be moved between warehouse locations to balance stock levels or route goods closer to the delivery destination.

3. Database Objectives

The SCMS database will serve as the single system of record enabling four classes of operational reports:

1. **Procurement reports:** supplier performance, purchase order aging, price validity tracking;
2. **Inventory reports:** stock levels, expiry alerts, zone utilization;
3. **Fulfillment reports:** order status, credit hold queue, FEFO allocation audit trail;
4. **Logistics reports:** shipment tracking, vehicle utilization, delivery success rates.

4. Identify Application Users

1. Procurement Manager

- Create/update a Supplier profile and record compliance audit score
- Add or revise a Product Catalog entry with negotiated price and validity dates
- Issue a Purchase Order (PO) to a supplier
- Blacklist a non-compliant vendor
- View PO status and pending deliveries

2. Warehouse Operator

- Record a Goods Receipt Note (GRN) against an incoming PO
- Reject a GRN (or partial batch) that fails quality inspection
- Assign a received batch to a specific warehouse zone
- Update inventory position after put-away
- Record a stock adjustment (damage, shrinkage)

3. Sales / Fulfillment Executive

- Create a new Customer profile with credit limit
- Place a Sales Order on behalf of a customer
- Check and enforce Credit Hold before approving an order
- Allocate inventory to an order using FEFO logic
- Process a customer return / cancellation

4. Logistics Coordinator

- Create a Shipment and group multiple orders into it
- Assign a vehicle (checking weight/volume capacity) to a shipment
- Assign a Delivery Agent to a shipment
- Update delivery status (In Transit → Delivered / Failed)
- Record a failed delivery and trigger re-dispatch

5. The Journey of a Good: Scenarios and Data Requirements

To understand the system's scope, let's follow the journey of a challenging product—a batch of temperature-sensitive vaccines—from the manufacturer to a local pharmacy.

Stage 1: Sourcing & Purchasing (The Procurement Challenge)

- **The Narrative:** Before we can sell the vaccines, the *Procurement Manager* needs a reliable vendor. Vendors frequently change their prices and lead times. The manager identifies a qualified supplier, negotiates a price valid for the next 6 months, and issues a Purchase Order (PO).
- **The Challenge:** Ordering from a blacklisted vendor, or losing track of agreed-upon pricing.
- **Data Captured for Database:** Supplier demographics, tax IDs, compliance audit scores, product catalog, negotiated pricing with validity dates, and Purchase Order records.

Stage 2: Arrival & Storage (The Warehousing Challenge)

- **The Narrative:** A truck arrives at the central warehouse. The *Warehouse Operator* inspects the goods. They aren't just receiving "vaccines"; they are receiving a specific physical batch manufactured on a certain date. Because these are temperature-sensitive, the operator must assign this specific batch to a specific "Cold Storage Zone" in the warehouse.
- **The Challenge:** Mixing up new stock with old stock, or storing perishables in an ambient temperature zone.
- **Data Captured for Database:** Goods Receipt Notes (GRN), Batch metadata (manufacturing/expiry dates, lot numbers), Warehouse profiles, Zone capacities, and the exact physical Inventory position (Product X from Batch Y is in Zone Z).
- **GRN Rejection Handling:** If a batch fails quality inspection (wrong temperature on arrival, damaged packaging, count mismatch), the operator records a partial or full rejection against the GRN. Rejected quantities are logged separately and do not enter available inventory. A rejection triggers a notification to the Procurement Manager to raise a supplier claim.

Stage 3: Customer Order & Allocation (The Fulfillment Challenge)

- **The Narrative:** A local retail pharmacy places an urgent order for the vaccines. The *Sales Executive* receives the order. First, the system checks if the pharmacy owes the company too much money (Credit Hold). Once approved, the system must figure out which specific vaccines to give them. It uses FEFO (First-Expired, First-Out) to automatically reserve the batch of vaccines that will expire soonest, saving the newer ones for later.
- **The Challenge:** Approving orders for defaulting customers, or shipping fresh stock while older stock rots in the warehouse.
- **Data Captured for Database:** Customer profiles, credit limits, Sales Order lines, and Inventory Allocation records (linking a specific order to a specific warehouse batch).
- **Returns & Cancellations:** If a customer returns goods or cancels after allocation, the system reverses the inventory allocation. Returned batches are quarantined and inspected before being re-added to available stock. The customer's outstanding balance is updated accordingly.

Stage 4: Shipping & Last-Mile (The Logistics Challenge)

- **The Narrative:** We have the vaccines packed in a box, but we have 50 other boxes going to the same city. The *Logistics Coordinator* groups these orders into a single "Shipment" to save money.

They assign the shipment to a refrigerated truck and a specific delivery agent. The agent drives to the pharmacy and records the final delivery status.

- **The Challenge:** Overloading a small van, or losing track of which driver has which customer's order.
- **Data Captured for Database:** Carrier contracts, Vehicle capacities (weight/volume), Shipment tracking states, and Delivery Agent assignments.

5. Key Queries the Database Must Answer

1. List all batches expiring within the next 30 days, sorted by expiry date (FEFO audit).
2. Find all Sales Orders currently on Credit Hold, with customer outstanding balance.
3. Show the complete inventory position — product, batch, zone, quantity available — for a given warehouse.
4. Retrieve all Purchase Orders raised against a specific supplier in the last 6 months, with delivery status.
5. Identify all GRNs that were partially or fully rejected, along with rejection reasons.
6. List all shipments currently "In Transit" with assigned driver and expected delivery date.
7. For a given Sales Order, trace the full chain: which batches were allocated, from which zone, on which shipment.
8. Find all suppliers whose compliance audit score falls below a threshold (blacklist candidates).
9. Calculate total inventory value per warehouse zone, grouped by product category.
10. Show all customers who have exceeded their credit limit in the last quarter.
11. List products that have fallen below their reorder threshold and have no open PO.
12. Report total weight and volume utilization per vehicle per shipment.